Calculus II – Spring 2025-26

Homework 2

Solution

1. Define each of the following terms. Give a full and complete mathematical definition. If your definition includes a secondary term that was defined in the course, you must define it as well! You do not have to define any term that appears prior to the words 'define the term'. You are not required to re-define terms that were already defined.

   Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers.

   (a) Define the term: the sequence $(a_n)_{n=1}^{\infty}$ is not bounded from below and not bounded from above, without using any negation word.

   Solution:

   We say that the sequence $(a_n)_{n=1}^{\infty}$ is not bounded from below and not bounded from above if for every $K_1 \in \mathbb{R}$ there exists an $n_1 \in \mathbb{N}$ for which $a_{n_1} < K_1$ and for every $K_2 \in \mathbb{R}$ there exists an $n_2 \in \mathbb{N}$ for which $a_{n_2} > K_2$.

   (b) Define the term: $\lim_{n\to\infty} a_n \neq \infty$, without using any negation word.

   Solution:

   We say that $\lim_{n\to\infty} a_n \neq \infty$ if there exists a $K > 0$ such that for every $N \in \mathbb{N}$ there exists an $n \geq N$ for which $a_n \leq K$.

   (c) Define the term: we have $\lim_{n\to\infty} a_n = -1$.

   Solution:

   We say that $\lim_{n\to\infty} a_n = -1$ if for every $\epsilon > 0$ there exists an $N \in \mathbb{N}$ such that for every $n \geq N$ we have $|a_n + 1| < \epsilon$.

   (d) Define the term: the sequence $(a_n)_{n=1}^{\infty}$ is strictly monotone.

   Solution:

   We say that the sequence $(a_n)_{n=1}^{\infty}$ is strictly monotone if for every $n \in \mathbb{N}$ we have $a_n > a_{n+1}$ or for every $n \in \mathbb{N}$ we have $a_n < a_{n+1}$.

2. Let $(a_n)_{n=1}^{\infty}$ be a sequence of real numbers.

Prove each of the following statements. In items (a) and (b) prove using only the definitions.

(a) For every $k \in \mathbb{N}$, the sequence $(a_{n+k})_{n=1}^{\infty}$ is convergent if and only if $(a_n)_{n=1}^{\infty}$ is convergent, and in that case, $\lim\limits_{n\to\infty} a_{n+k} = \lim\limits_{n\to\infty} a_n$.

Solution:

Let $k \in \mathbb{N}$.

We define $(b_n)_{n=1}^{\infty} = (a_{n+k})_{n=1}^{\infty}$.

$\underline{\Leftarrow:}$ As $(a_n)_{n=1}^{\infty}$ is convergent, then there exists an $L \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} a_n = L$.

Let $\epsilon > 0$. As $\lim\limits_{n\to\infty} a_n = L$, then there exists an $N_1 \in \mathbb{N}$ such that for every $n \geq N_1$ we have

$$|a_n - L| < \boxed{\epsilon} \qquad (\star)$$

We choose $\boxed{N = N_1}$. Let $n \geq N$, therefore $n + k \geq N_1$ and therefore $(\star)$ holds for $n + k$.

Then

$$|b_n - L| = |a_{n+k} - L| < \boxed{\epsilon}$$

Thus $(b_n)_{n=k}^{\infty}$ is convergent and $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = L$.

$\underline{\Rightarrow:}$ As $(b_n)_{n=1}^{\infty}$ is convergent, then there exists an $L \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} b_n = L$.

Let $\epsilon > 0$. As $\lim\limits_{n\to\infty} b_n = L$, then there exists an $N_1 \in \mathbb{N}$ such that for every $n \geq N_1$ we have

$$|b_n - L| < \boxed{\epsilon} \qquad (\star\star)$$

We choose $\boxed{N = N_1 + k}$. Let $n \geq N$, therefore $n - k \geq N_1$ and therefore $(\star\star)$ holds for $n - k$.

Then

$$|a_n - L| = |b_{n-k} - L| < \boxed{\epsilon}$$

Thus $(a_n)_{n=1}^{\infty}$ is convergent and $\lim\limits_{n\to\infty} b_n = \lim\limits_{n\to\infty} a_n = L$.

(b) If $(a_n)_{n=1}^{\infty}$ is increasing, then $(a_n)_{n=1}^{\infty}$ converges in the extended sense and

$$\lim_{n\to\infty} a_n = \sup\{a_n : n \in \mathbb{N}\}$$

Solution:

First we assume that $(a_n)_{n=1}^{\infty}$ is bounded from above.

As $(a_n)_{n=1}^{\infty}$ is bounded from above (and clearly not empty), then by the least upper bound theorem, there exists an $\bar{s} \in \mathbb{R}$ such that $\bar{s} = \sup\{a_n \mid n \in \mathbb{N}\}$.

Let $\epsilon > 0$. By the ϵ property of $\bar{s}$, there exists an $n_1 \geq 1$ for which $\bar{s} - \epsilon < a_{n_1} \leq \bar{s}$.

We choose $N \in \mathbb{N}$ such that $N > n_1$ (by the Archimedean property there exists such N). let $n \geq N$.

As $(a_n)_{n=1}^{\infty}$ is increasing and $n > n_1$, we have $\bar{s} - \epsilon < a_{n_1} \leq a_n$ note that $a_n \leq \bar{s} < \bar{s} + \epsilon$, so

$$\bar{s} - \epsilon < a_{n_1} \leq a_n < \bar{s} + \epsilon$$

therefore $|a_n - \bar{s}| < \epsilon$. We proved that $(a_n)_{n=1}^{\infty}$ converges, and that $\lim\limits_{n\to\infty} a_n = \bar{s}$.

Now we assume that $(a_n)_{n=1}^{\infty}$ is not bounded from above.

Let $K > 0$. As $(a_n)_{n=1}^{\infty}$ is not bounded from above, there exists an $n_1 \geq 1$ for which $a_{n_1} > K$.

We choose $N \in \mathbb{N}$ such that $N > n_1$ (by the Archimedean property there exists such N). Let $n \geq N$.

As $(a_n)_{n=1}^{\infty}$ is increasing and $n > n_1$, we have $a_n \geq a_{n_1} > K$, thus $\lim\limits_{n\to\infty} a_n = \sup\{a_n \mid n \in \mathbb{N}\} = \infty$.

(c) Suppose that $(a_n)_{n=1}^{\infty}$ is bounded and consider the sequence $(b_k)_{k=1}^{\infty}$ defined by

$$b_k = \inf\{a_n : n \geq k\} \qquad \forall k \in \mathbb{N}$$

Prove that $(b_k)_{k=1}^{\infty}$ is convergent.

Hint: prove that $(b_k)_{k=1}^{\infty}$ is increasing and bounded from above.

Solution:

First, as the sequence $(a_n)_{n=1}^{\infty}$ is bounded, then for every $k \in \mathbb{N}$, the set $\{a_n : n \geq k\}$ is bounded from below, and as it is not empty, then by the greatest lower bound theorem b_k is well defined.

Also, as $(a_n)_{n=1}^{\infty}$ is bounded from above, then there exists an $M \in \mathbb{R}$ such that $a_n \leq M$ for every $n \in \mathbb{N}$. It follows that for every $k \in \mathbb{N}$, M is an upper bound of the set $\{a_n : n \geq k\}$ and therefore $b_k = \inf\{a_n : n \geq k\} \leq M$ for every $k \in \mathbb{N}$. Thus, the sequence $(b_k)_{k=1}^{\infty}$ is bounded from above.

We now show that $(b_k)_{k=1}^{\infty}$ is increasing.

Let $k \in \mathbb{N}$. Note that $\{a_n : n \geq k+1\} \subseteq \{a_n : n \geq k\}$ and therefore

$b_{k+1} = \inf\{a_n : n \geq k+1\} \geq \inf\{a_n : n \geq k\} = b_k$.

It follows that the sequence $(b_k)_{k=1}^{\infty}$ is increasing and bounded from above, and therefore is convergent.

3. (a) Let $a \in \mathbb{R}$ and let f be a function that is defined on $[a, \infty)$. Assume that $\lim_{x\to\infty} f(x) = \infty$.

Prove that for every sequence $(x_n)_{n=1}^{\infty}$ such that $\lim_{n\to\infty} x_n = \infty$, we have $\lim_{n\to\infty} f(x_n) = \infty$.

Solution:

Let $(x_n)_{n=1}^{\infty}$ be a sequence such that $\lim_{n\to\infty} x_n = \infty$.

Let $K > 0$. As $\lim_{x\to\infty} f(x) = \infty$ then there exists an $M_1 > 0$ such that for every $x > M_1$ we have

$$f(x) > \boxed{K} \qquad (\star)$$

As $\lim_{n\to\infty} x_n = \infty$ then there exists an $N_1 \in \mathbb{N}$ such that for every $n \geq N_1$ we have

$$x_n > \boxed{M_1} \qquad (\star\star)$$

Choose $N = N_1$ and let $n \geq N$. Then $(\star\star)$ holds for n and therefore $(\star)$ holds for x_n. It follows that

$$f(x_n) \overset{(\star)}{>} \boxed{K}$$

therefore $\lim_{n\to\infty} f(x_n) = \infty$.

(b) Compute the following limit:

$$\lim_{n\to\infty} \left[n + n^2 \ln n - n^2 \ln(n+1)\right]$$

Solution:

We have

$$\lim_{n\to\infty} \left[n + n^2 \ln n - n^2 \ln(n+1)\right] = \lim_{n\to\infty} \left[n - n^2 \ln\left(\frac{n+1}{n}\right)\right] = \lim_{n\to\infty} n^2 \left[\frac{1}{n} - \ln\left(\frac{n+1}{n}\right)\right]$$

$$= \lim_{n\to\infty} \frac{\frac{1}{n} - \ln\left(1+\frac{1}{n}\right)}{\frac{1}{n^2}} \overset{x_n = \frac{1}{n}}{=} \lim_{x\to 0} \frac{x - \ln(1+x)}{x^2} \overset{\frac{0}{0}}{=} \lim_{x\to 0} \frac{1 - \frac{1}{1+x}}{2x}$$

$$\overset{\frac{0}{0}}{=} \lim_{x\to 0} \frac{\frac{1}{(1+x)^2}}{2} = \frac{1}{2}$$

4. Compute the following limit as a function of $0 < \alpha < 1$:

$$\lim_{n\to\infty}\left[(n+1)^{2\alpha}-\left(n^2+1\right)^{\alpha}\right]$$

Solution:

$$\lim_{n\to\infty}\left((n+1)^{2\alpha}-(n^2+1)^{\alpha}\right) \overset{x_n=n}{=} \lim_{x\to\infty}\left((x+1)^{2\alpha}-(x^2+1)^{\alpha}\right) = \lim_{x\to\infty} x^{2\alpha}\left[\left(1+\frac{1}{x}\right)^{2\alpha}-\left(1+\frac{1}{x^2}\right)^{\alpha}\right] =$$

$$\lim_{x\to\infty}\frac{\left(1+\frac{1}{x}\right)^{2\alpha}-\left(1+\frac{1}{x^2}\right)^{\alpha}}{x^{-2\alpha}} \overset{L}{=} \lim_{x\to\infty}\frac{-\frac{2\alpha}{x^2}\left(1+\frac{1}{x}\right)^{2\alpha-1}+\frac{2\alpha}{x^3}\left(1+\frac{1}{x^2}\right)^{\alpha-1}}{-2\alpha x^{-2\alpha-1}} = \lim_{x\to\infty}\frac{x^{-2}\left(1+\frac{1}{x}\right)^{2\alpha-1}-x^{-3}\left(1+\frac{1}{x^2}\right)^{\alpha-1}}{x^{-2\alpha-1}} =$$

$$\lim_{x\to\infty}\left[x^{2\alpha-1}\left(1+\frac{1}{x}\right)^{2\alpha-1}-x^{2\alpha-2}\left(1+\frac{1}{x^2}\right)^{\alpha-1}\right]$$

Now we consider three cases.

<u>**Case 1:**</u> $0 < \alpha < \frac{1}{2}$

$$\lim_{x\to\infty}\left[x^{2\alpha-1}\left(1+\frac{1}{x}\right)^{2\alpha-1}-x^{2\alpha-2}\left(1+\frac{1}{x^2}\right)^{\alpha-1}\right] \overset{\text{AOL}}{=} 0-0=0$$

<u>**Case 2:**</u> $\alpha = \frac{1}{2}$

$$\lim_{x\to\infty}\left[x^{2\alpha-1}\left(1+\frac{1}{x}\right)^{2\alpha-1}-x^{2\alpha-2}\left(1+\frac{1}{x^2}\right)^{\alpha-1}\right] \overset{\text{AOL}}{=} 1-0=0$$

<u>**Case 3:**</u> $\alpha > \frac{1}{2}$

$$\lim_{x\to\infty}\left[x^{2\alpha-1}\left(1+\frac{1}{x}\right)^{2\alpha-1}-x^{2\alpha-2}\left(1+\frac{1}{x^2}\right)^{\alpha-1}\right] \overset{\text{AOL}}{=} \infty-0=0$$

We conclude that

$$\lim_{n\to\infty}\left[(n+1)^{2\alpha}-\left(n^2+1\right)^{\alpha}\right] = \begin{cases} 0 & 0<\alpha<\frac{1}{2} \\ 1 & \alpha=\frac{1}{2} \\ \infty & \alpha>\frac{1}{2} \end{cases}$$

5. (a) Consider the sequence $(a_n)_{n=1}^{\infty}$ defined by the recursive formula

$$\begin{cases} a_{n+1} = a_n + \frac{1}{a_n} & \forall n \in \mathbb{N} \\ a_1 = 1 \end{cases}$$

Prove that $\lim_{n\to\infty} a_n = \infty$.

Solution:

We first prove that $\forall n \in \mathbb{N}$ we have $a_n > 0$, and therefore the sequence $(a_n)_{n=1}^{\infty}$ is well-defined.

We prove this claim by induction:

<u>Base case</u>: $a_1 > 0$.

<u>Induction step</u>: Let $n \in \mathbb{N}$ such that $a_n > 0$. Then $a_{n+1} = a_n + \frac{1}{a_n} > 0$.

We now prove that the sequence $(a_n)_{n=1}^{\infty}$ is increasing. Let $n \in \mathbb{N}$. Then

$$a_{n+1} = a_n + \frac{1}{a_n} > a_n$$

ATC that $\lim_{n\to\infty} a_n \neq \infty$. As $(a_n)_{n=1}^{\infty}$ is increasing, then there exists an $L \in \mathbb{R}$ such that

$$L = \lim_{n\to\infty} a_n \overset{\text{increasing}}{=} \sup\{a_n \,:\, n \in \mathbb{N}\}.$$

Note that $L \overset{\text{upper bound}}{\geq} a_1 = 1$ and therefore $L \neq 0$.

It follows that

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty}\left[a_n + \frac{1}{a_n}\right] \overset{\text{AOL}}{=} L + \frac{1}{L}$$

Thus $\frac{1}{L} = 0$. Contradiction.

(b) i. Let $0 \le \alpha \le 2$. Prove that $\alpha \le \sqrt{2\alpha} \le 2$.

Solution:

We first prove the right inequality. As $0 \le \alpha \le 2$, then we have $\sqrt{\alpha} \le \sqrt{2}$.

Multiplying both sides of the inequality by $\sqrt{2}$ yields $\sqrt{2\alpha} \le \sqrt{2}^2 = 2$.

We now prove the left inequality. As $0 \le \alpha \le 2$, then $\alpha^2 \le 2\alpha$.

Taking the square root to both sides of the inequality yields $\alpha \le \sqrt{2\alpha}$.

ii. Let $(a_n)_{n=1}^{\infty}$ be the sequence of real numbers defined by the recursive formula

$$\begin{cases} a_{n+1} = \sqrt{2a_n} & \forall n \in \mathbb{N} \\ a_1 = \sqrt{2} \end{cases}$$

Prove that the sequence $(a_n)_{n=1}^{\infty}$ is convergent.

Solution:

We first prove that for every $n \in \mathbb{N}$ we have $a_n \ge 0$, from which follows that the sequence $(a_n)_{n=1}^{\infty}$ is well-defined. We prove that by using induction:

<u>Base case</u>: we have $a_1 = \sqrt{2} \ge 0$.

<u>Induction step</u>: Let $n \in \mathbb{N}$ and suppose that $a_n \ge 0$. Then

$$a_{n+1} = \sqrt{2a_n} \overset{a_n \ge 0}{\ge} 0$$

We now prove that the sequence $(a_n)_{n=1}^{\infty}$ is increasing and bounded from above by 2, from which follows that the sequence $(a_n)_{n=1}^{\infty}$ is convergent. We prove that for every $n \in \mathbb{N}$ we have $a_n \le a_{n+1} \le 2$ by induction.

<u>Base case</u>: we have $a_1 = \sqrt{2} \overset{\sqrt{2} \ge 1}{\le} \sqrt{2 \cdot \sqrt{2}} = a_2$.

<u>Induction step</u>: Let $n \in \mathbb{N}$ and suppose that $a_n \le a_{n+1} \le 2$. As $0 \le a_{n+1} \le 2$, then by the previous item we have $a_{n+1} \le \sqrt{2 \cdot a_{n+1}} \le 2 \Rightarrow a_{n+1} \le a_{n+2} \le 2$.

iii. Prove that $\sqrt{2\sqrt{2\sqrt{2\sqrt{2\ldots}}}} = 2$, that is, prove that $\lim\limits_{n\to\infty} a_n = 2$.

Solution:

As the sequence $(a_n)_{n=1}^{\infty}$ is convergent, then there exists an $L \in \mathbb{R}$ such that $L = \lim\limits_{n\to\infty} a_n \overset{\text{increasing}}{=} \sup\{a_n : n \in \mathbb{N}\}$.

Note that $L \overset{\text{upper bound}}{\ge} a_1 = \sqrt{2}$ and therefore $L \ne 0$. We now compute

$$L = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \sqrt{2a_n} \overset{\text{AOL}}{=} \sqrt{2L}$$

Thus $L = \sqrt{2L}$ and therefore $L^2 = 2L$ and therefore $L = 0$ or $L = 2$. As $L \ne 0$ then $L = 2$.

6. Let $0 < \beta < \alpha$ and let $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ be two sequences that are defined by the recursive formulas

$$\begin{cases} a_{n+1} = \frac{a_n+b_n}{2}, & b_{n+1} = \frac{2a_nb_n}{a_n+b_n}, \qquad \forall n \in \mathbb{N} \\ a_1 = \alpha, & b_1 = \beta \end{cases}$$

(a) Prove that $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are well-defined, and prove that $a_{n+1} \geq b_{n+1}$ for every $n \in \mathbb{N}$.

Solution:

We first prove by induction that $a_n, b_n > 0$ for every $n \in \mathbb{N}$.

Base case: $(n = 1)$. It is already given that $0 < b_1 < a_1$.

Induction step: $(n \Rightarrow n+1.)$ Let $n \in \mathbb{N}$ and suppose that $a_n, b_n > 0$. Then $a_{n+1} = \frac{a_n+b_n}{2} > 0$, $b_{n+1} = \frac{2a_nb_n}{a_n+b_n} > 0$.

Thus $(a_n)_{n=1}^{\infty}$ and $(b_n)_{n=1}^{\infty}$ are well-defined.

We prove now that $a_{n+1} \geq b_{n+1}$ for every $n \in \mathbb{N}$. Let $n \in \mathbb{N}$.

As $(a_n - b_n)^2 \geq 0$ for every $n \in \mathbb{N}$, then

$$a_n^2 - 2a_nb_n + b_n^2 \geq 0 \Rightarrow a_n^2 + 2a_nb_n - 4a_nb_n + b_n^2 \geq 0 \Rightarrow a_n^2 + 2a_nb_n + b_n^2 \geq 4a_nb_n \Rightarrow (a_n + b_n) \cdot (a_n + b_n) \geq 4a_nb_n$$

We divide both sides by $2(a_n + b_n)$ and we obtain

$$a_{n+1} = \frac{a_n + b_n}{2} \geq \frac{2a_nb_n}{a_n + b_n} = b_{n+1}.$$

(b) Prove that there exists an $L \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} a_n = \lim\limits_{n\to\infty} b_n = L$.

Solution:

We prove that $(a_n)_{n=1}^{\infty}$ is decreasing.

Let $n \in \mathbb{N}$. We note that

$$a_{n+1} = \frac{a_n + b_n}{2} \leq \frac{a_n + a_n}{2} = a_n$$

We conclude that $a_n \leq a_1$ for every $n \in \mathbb{N}$.

We prove that $(b_n)_{n=1}^{\infty}$ is increasing.

Let $n \in \mathbb{N}$. We note that

$$b_{n+1} = \frac{2a_nb_n}{a_n + b_n} \geq \frac{2a_nb_n}{a_n + a_n} = b_n$$

We conclude that $b_n \geq b_1$ for every $n \in \mathbb{N}$.

Thus for every $n \in \mathbb{N}$ we have

$$b_1 \leq b_n \leq a_n \leq a_1$$

Thus $(a_n)_{n=1}^{\infty}$ is bounded from below by b_1 and $(b_n)_{n=1}^{\infty}$ is bounded from above by a_1 and therefore there exist $L_1, L_2 \in \mathbb{R}$ such that $\lim\limits_{n\to\infty} a_n = L_1$ and $\lim\limits_{n\to\infty} b_n = L_2$.

We now show $L_1 = L_2$ and complete the proof:

$$L_1 = \lim_{n\to\infty} a_{n+1} = \lim_{n\to\infty} \left(\frac{a_n + b_n}{2}\right) = \frac{L_1 + L_2}{2} \Rightarrow \frac{L_1}{2} = \frac{L_2}{2} \Rightarrow L_1 = L_2.$$

(c) Prove that $a_n b_n = a_1 b_1$ for every $n \in \mathbb{N}$ and find the value of L.

Solution:

We prove by induction that $a_n b_n = a_1 b_1$ for every $n \in \mathbb{N}$.

Base case: Clearly $a_1 b_1 = a_1 b_1$.

Induction step: Let $n \in \mathbb{N}$. We assume that $a_n b_n = a_1 b_1$. Then $a_{n+1} b_{n+1} = \frac{a_n + b_n}{2} \cdot \frac{2a_n b_n}{a_n + b_n} = a_n b_n = a_1 b_1$.

Now we find the value of L.

We note that $\lim_{n\to\infty} [a_1 b_1] = a_1 b_1$.

By item (b) we obtain that $\lim_{n\to\infty} [a_n b_n] = L^2$

Thus $L^2 = a_1 b_1 \Rightarrow L = \pm\sqrt{a_1 b_1}$

As $a_n, b_n > 0$ for every $n \in \mathbb{N}$, then by the monotonicity of limits we have $L \geq 0$, thus $L = \sqrt{a_1 b_1}$.